New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

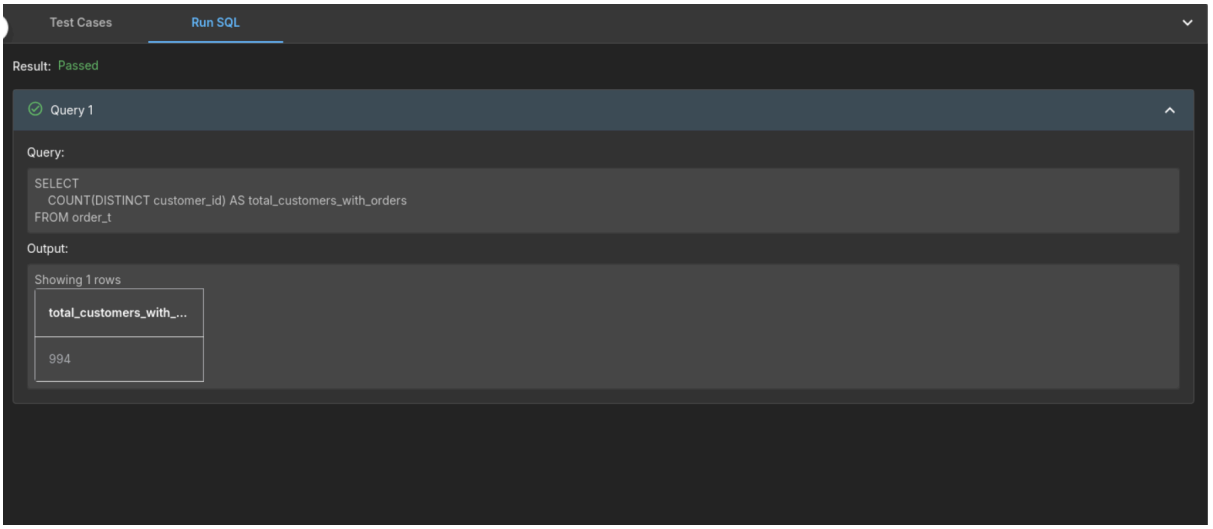
Business Questions

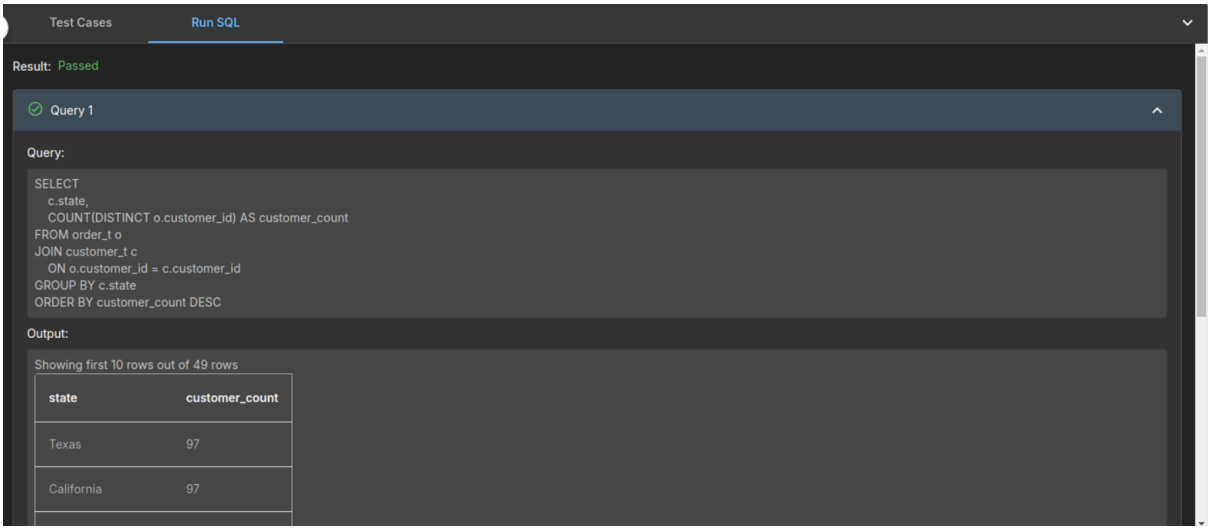
Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
-- Total number of customers who have placed at least one order
SELECT
    COUNT(DISTINCT customer_id) AS total_customers_with_orders
FROM order_t;
-- Distribution of customers across states (only customers who placed orders)
SELECT
    c.state,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:


Test Cases
Run SQL
Result: Passed
Query 1
Query:
SELECT
COUNT(DISTINCT customer_id) AS total_customers_with_orders
FROM order_t
Output:
Showing 1 rows
total_customers_with_...
994


Test Cases
Run SQL
Result: Passed
Query 1
Query:
SELECT
c.state,
COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN customer_t c
ON o.customer_id = c.customer_id
GROUP BY c.state
ORDER BY customer_count DESC
Output:
Showing first 10 rows out of 49 rows
state
customer_count
Texas
97
California
97


Florida
86
New York
69
District of Columbia
35
Ohio
33
Colorado
33
Alabama
29
Washington
28
Arizona
26

Observations and Insights:

Insight 1: The active customer base is relatively small and concentrated

- Only 994 unique customers have placed orders.
- For a vehicle resale platform operating nationally, this is a thin buyer base, not a mass-market footprint.

Observation:
Growth problems aren't subtle here. When sales dip, the impact is magnified because the customer pool itself is limited. This makes retention and experience quality disproportionately important.

Insight 2: Demand is highly concentrated in a few large states

- Texas (97) and California (97) jointly contribute the highest number of ordering customers.
- Florida (86) and New York (69) follow, then a sharp drop.

Observation:
A small number of populous states dominate demand. This suggests:

- Marketing and logistics are likely most effective in these regions
- Any operational failure (shipping delays, bad after-sales experience) in these states would disproportionately hurt revenue and online ratings

This is concentration risk, not diversification.

Insight 3: Long tail of states with low customer penetration

- There are 49 states/regions represented, but most have very low customer counts.
- Many states contribute fewer than 30 customers.

Observation:
This is not organic nationwide adoption — it's shallow geographic reach. Either:

- Awareness is weak outside key states, or
- Shipping/service quality discourages repeat or regional expansion

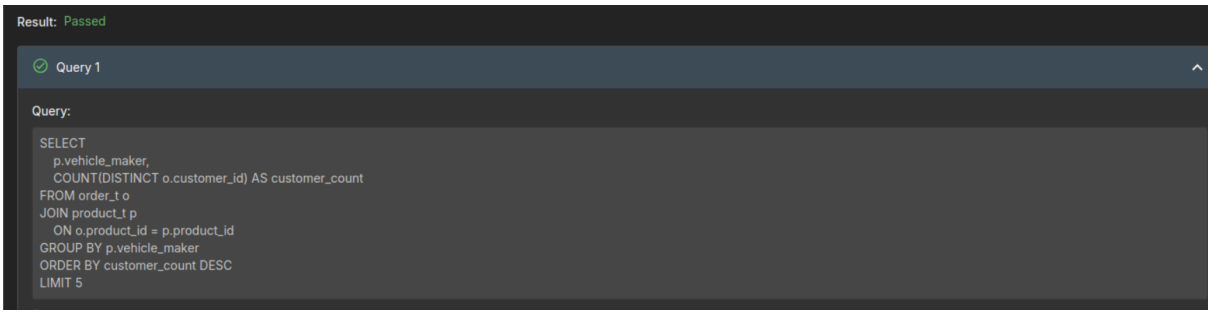
Both explanations align with declining ratings and quarterly customer drop-off.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM order_t o
JOIN product_t p
    ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

Output:





Observations and Insights:

Insight 1: Customer preference is skewed toward a few legacy brands

Chevrolet leads with 83 customers, followed by Ford (63) and Toyota (52).

Observation:

Customer preference is concentrated around well-established, mass-market brands. This suggests that in the pre-owned vehicle space, brand familiarity outweighs novelty or premium positioning, especially when trust and resale value matter.

Insight 2: Preference drop-off after the top brand is significant

There is a clear gap between Chevrolet and Ford, and another tapering beyond Toyota.

Observation:

Chevrolet's lead indicates a disproportionate pull, not a marginal one. Any disruption in sourcing, pricing, or post-sale service for this brand would materially affect customer demand and overall sales volume.

Insight 3: Tight clustering among ranks 3–5 indicates weak brand differentiation

Toyota, Pontiac, and Dodge are tightly clustered around 50–52 customers.

Observation:

Beyond the top two brands, customer preference does not strongly favor one maker over another. This implies limited differentiation in the resale experience, shifting competitive advantage away from brand and toward pricing, availability, shipping reliability, and after-sales service.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
    state,
    vehicle_maker,
    customer_count
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT o.customer_id) AS customer_count,
        RANK() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(DISTINCT o.customer_id) DESC
        ) AS rnk
    FROM order_t o
    JOIN customer_t c
        ON o.customer_id = c.customer_id
    JOIN product_t p
        ON o.product_id = p.product_id
    GROUP BY
        c.state,
        p.vehicle_maker
) ranked_makers
WHERE rnk = 1
```

ORDER BY state;

Output:

Query 1

Query:

```
SELECT
    state,
    vehicle_maker,
    customer_count
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT o.customer_id) AS customer_count,
        RANK() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(DISTINCT o.customer_id) DESC
        ) AS rnk
    FROM order_t o
    JOIN customer_t c
        ON o.customer_id = c.customer_id
    JOIN product_t p
        ON o.product_id = p.product_id
    GROUP BY
        c.state,
        p.vehicle_maker
) ranked_makers
WHERE rnk = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1

Observations and Insights:

Insight 1: Vehicle maker preference is highly regional, not national

Different states show different top vehicle makers, and the nationally preferred brands do not dominate every state.

Observation:

Customer preference varies significantly by geography. This indicates that a single nationwide inventory or marketing strategy will underperform compared to state-level or region-level customization.

Insight 2: Many states show ties for the most preferred vehicle maker

Several states (e.g., Arizona, Arkansas) have multiple vehicle makers sharing rank 1, often with very small customer counts.

Observation:

In these states, preference is weakly formed and highly sensitive to small changes in availability, pricing, or delivery experience. This suggests immature or low-volume markets rather than strong brand loyalty.

Insight 3: Customer counts per preferred maker are generally low at the state level

Even the most preferred vehicle makers in many states have single-digit customer counts.

Observation:

State-level demand is thin across most regions. This reinforces the earlier finding of limited customer penetration and explains why negative feedback or service issues can disproportionately impact quarterly performance and ratings.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
--The overall average rating query first

SELECT

   AVG(rating) AS overall_avg_rating

FROM (

   SELECT

      CASE customer_feedback

         WHEN 'Very Bad' THEN 1

         WHEN 'Bad' THEN 2
```

```
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
            ELSE NULL
        END AS rating
    FROM order_t
) feedback_numeric;
-- The quarterly average rating query
SELECT
    quarter_number,
    AVG(rating) AS avg_rating_per_quarter
FROM (
    SELECT
        quarter_number,
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
            ELSE NULL
        END AS rating
    FROM order_t
) feedback_numeric
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Query 1

Query:

```
SELECT
   AVG(rating) AS overall_avg_rating
FROM (
   SELECT
      CASE customer_feedback
         WHEN 'Very Bad' THEN 1
         WHEN 'Bad' THEN 2
         WHEN 'Okay' THEN 3
         WHEN 'Good' THEN 4
         WHEN 'Very Good' THEN 5
         ELSE NULL
      END AS rating
   FROM order_t
) feedback_numeric
```

Output:

Showing 1 rows

overall_avg_rating
3.135

Result: Passed

Query 1

Query:

```
SELECT
   quarter_number,
   AVG(rating) AS avg_rating_per_quarter
FROM (
   SELECT
      quarter_number,
      CASE customer_feedback
         WHEN 'Very Bad' THEN 1
         WHEN 'Bad' THEN 2
         WHEN 'Okay' THEN 3
         WHEN 'Good' THEN 4
         WHEN 'Very Good' THEN 5
         ELSE NULL
      END AS rating
   FROM order_t
) feedback_numeric
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Output:

Showing 4 rows

quarter_number	avg_rating_per_quarter
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.396984924623156

Observations and Insights:

Insight 1: Overall customer satisfaction is below “Good”

An overall average rating of 3.135 sits just above “Okay” and well below “Good” (4).

Observation:

Customer sentiment is mediocre at best. This indicates that while customers are not universally unhappy, the experience is not strong enough to drive repeat purchases or positive word-of-mouth, which aligns with declining sales and poor online ratings.

Insight 2: Customer satisfaction has declined consistently quarter over quarter

Ratings fall every single quarter, from 3.55 in Q1 to 2.40 in Q4.

Observation:

This is a systemic deterioration, not a one-off issue. It suggests worsening performance in areas such as shipping timelines, after-sales service, or issue resolution, rather than isolated incidents.

Insight 3: Customer sentiment crossed into negative territory by Q4

By Q3, the average rating drops below 3 (“Okay”), and by Q4 it approaches “Bad”.

Observation:

Once sentiment falls below neutral, customer trust erodes rapidly. At this stage, acquiring new customers becomes harder, refunds and complaints increase, and negative reviews begin to outweigh positive ones — explaining the sharp drop in new customers noted by leadership.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
-- Quarterly percentage distribution

SELECT

    CAST(quarter_number AS VARCHAR) AS period,

    COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*) AS very_bad_pct,

    COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*) AS bad_pct,

    COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*) AS okay_pct,

    COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*) AS good_pct,

    COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*) AS very_good_pct

FROM order_t

GROUP BY quarter_number

UNION ALL

-- Overall percentage distribution

SELECT

    'Overall' AS period,

    COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*) AS very_bad_pct,

    COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*) AS bad_pct,

    COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*) AS okay_pct,
```

COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*) AS good_pct,

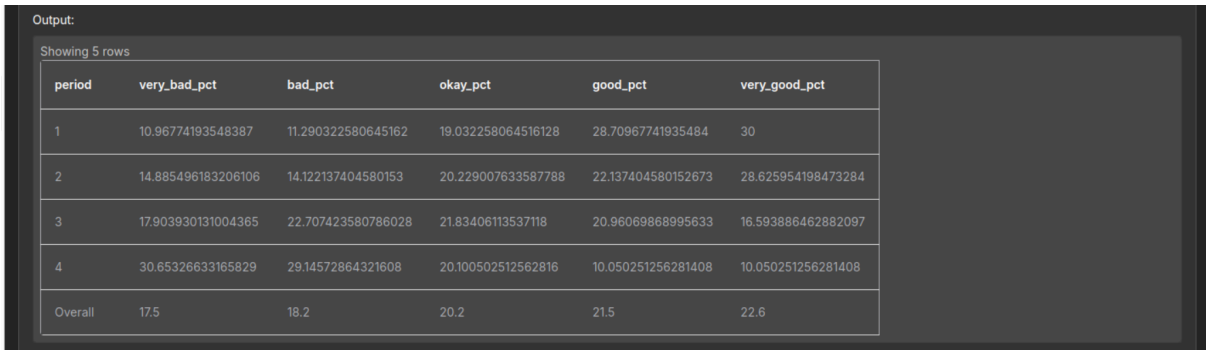
COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*) AS very_good_pct

FROM order_t

ORDER BY period;

Output:





Observations and Insights:

Insight 1: Customer dissatisfaction increases sharply over time

The combined percentage of negative feedback (“Bad” + “Very Bad”) rises every quarter, peaking at nearly 60% in Q4.

Observation:

This confirms that customers are becoming progressively more dissatisfied, not sporadically unhappy. The dissatisfaction trend is structural and worsening, which strongly explains declining sales and deteriorating online ratings.

Insight 2: Positive customer experiences collapse in later quarters

Positive feedback (“Good” + “Very Good”) drops from ~59% in Q1 to just ~20% in Q4.

Observation:

This indicates a severe breakdown in the customer experience. Even satisfied customers in earlier quarters are no longer having positive outcomes, signaling failures in execution rather than expectation management.

Insight 3: Neutral feedback converts into negative feedback

The percentage of “Okay” feedback remains relatively stable across quarters (~20%), while negative feedback surges.

Observation:

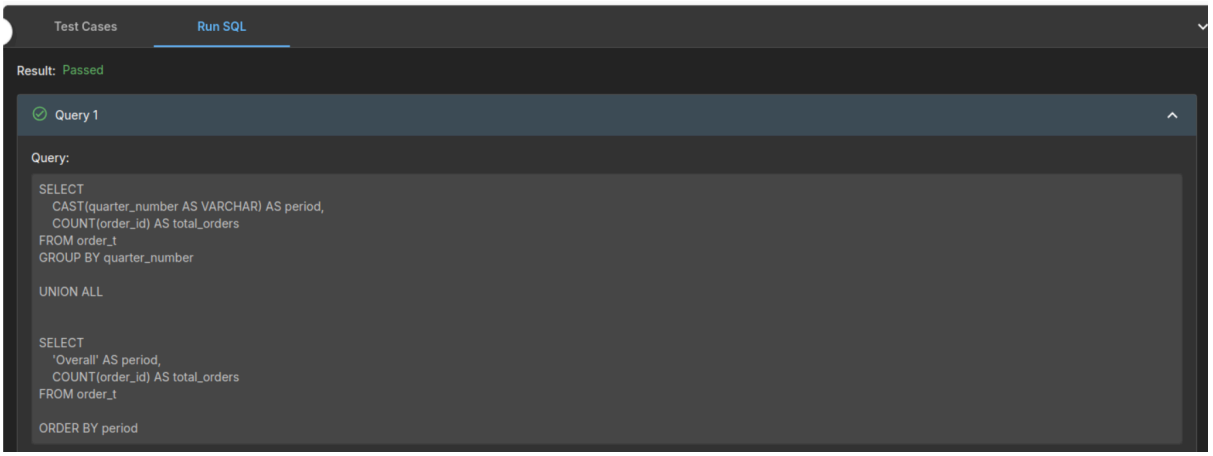
Customers are not staying neutral — they are tipping into dissatisfaction. This is a classic churn precursor: once neutral customers turn negative, recovery becomes exponentially harder and more expensive.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    CAST(quarter_number AS VARCHAR) AS period,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
UNION ALL
-- Overall total orders
SELECT
    'Overall' AS period,
    COUNT(order_id) AS total_orders
FROM order_t
ORDER BY period;
```

Output:





Observations and Insights:

Insight 1: Orders decline consistently quarter over quarter
Order volume drops every quarter, from 310 in Q1 to 199 in Q4, representing a decline of ~36% over the year.
Observation:
This is not seasonal noise. The steady downward trajectory indicates a structural issue in demand or customer experience rather than short-term market fluctuation.

Insight 2: The steepest decline aligns with worsening customer sentiment

The most pronounced reductions occur after Q2, coinciding with the quarters where customer ratings and negative feedback increased sharply.

Observation:

This temporal alignment suggests a strong relationship between deteriorating customer satisfaction and reduced order volumes, reinforcing that poor experience is directly impacting sales.

Insight 3: Overall order volume masks accelerating late-stage decline

While the total number of orders for the year is 1000, nearly 31% of these occurred in Q1 alone.

Observation:

Relying on annual totals would obscure the severity of recent performance deterioration. The business is entering a weaker baseline going into the next year, increasing risk if corrective actions are not taken promptly.

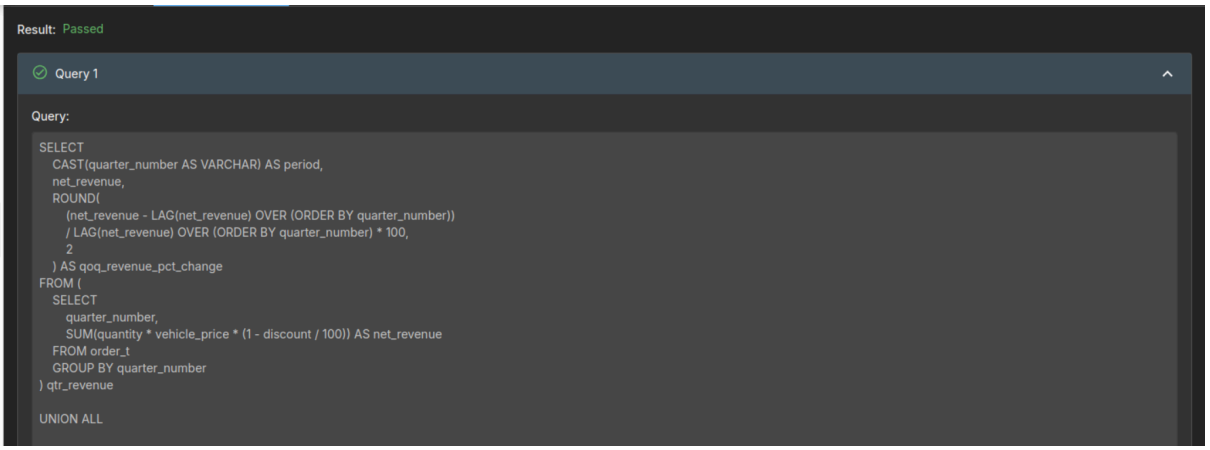
Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

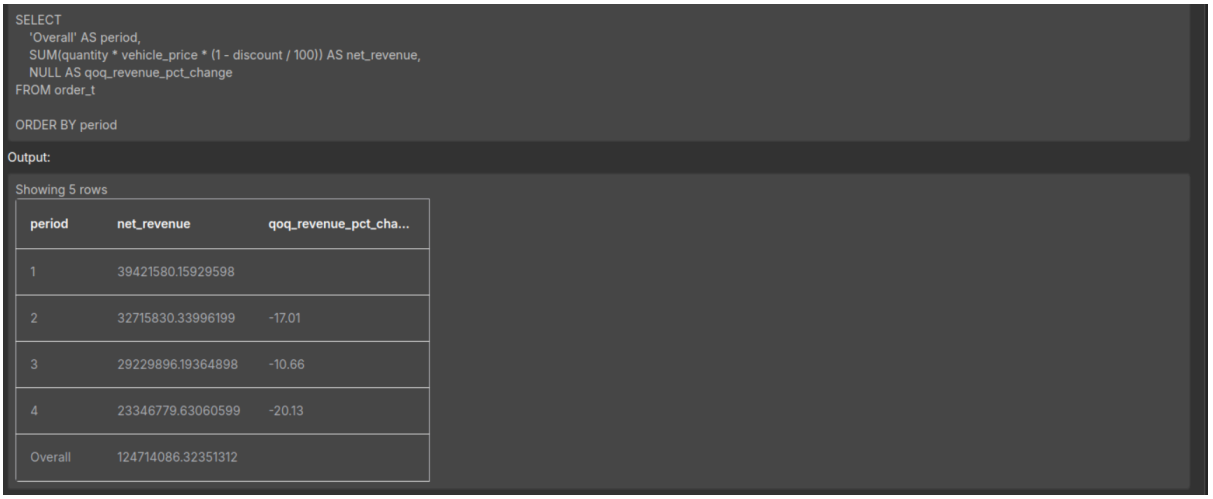
Solution Query:

```
-- Quarterly net revenue with QoQ % change
SELECT
    CAST(quarter_number AS VARCHAR) AS period,
    net_revenue,
    ROUND(
        (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
        / LAG(net_revenue) OVER (ORDER BY quarter_number) * 100,
        2
    ) AS qoq_revenue_pct_change
FROM (
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 - discount / 100)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) qtr_revenue
UNION ALL
-- Overall net revenue (summary row)
SELECT
    'Overall' AS period,
    SUM(quantity * vehicle_price * (1 - discount / 100)) AS net_revenue,
    NULL AS qoq_revenue_pct_change
FROM order_t
```

ORDER BY period;

Output:





Observations and Insights:

Insight 1: Net revenue shows a consistent downward trend

Observation:

Net revenue declines every quarter from Q1 to Q4, with no quarter showing recovery or stabilization. The company is experiencing sustained revenue erosion, indicating systemic business issues rather than seasonal or one-off fluctuations. Immediate corrective action is required to prevent further financial deterioration.

Insight 2: Revenue decline accelerates in the final quarter

Observation:

The steepest quarter-over-quarter revenue drop occurs in Q4 (–20.13%), which is significantly higher than the decline in Q3. This acceleration suggests compounding

negative effects such as worsening customer satisfaction, reduced repeat purchases, and declining brand trust, which intensify over time if not addressed early.

Insight 3: Discounts are not preventing revenue loss

Observation:

Despite discounts being applied (factored into net revenue), revenue continues to decline sharply each quarter. Price reductions alone are ineffective in sustaining revenue. The decline is likely driven by non-pricing factors such as after-sales service quality, shipping experience, and overall customer dissatisfaction.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
-- Quarterly trend
SELECT
    CAST(quarter_number AS VARCHAR) AS period,
    COUNT(order_id) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount / 100)) AS net_revenue
FROM order_t
GROUP BY quarter_number
UNION ALL
-- Overall summary row
SELECT
    'Overall' AS period,
    COUNT(order_id) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount / 100)) AS net_revenue
FROM order_t
ORDER BY period;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    CAST(quarter_number AS VARCHAR) AS period,
    COUNT(order_id) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount / 100)) AS net_revenue
FROM order_t
GROUP BY quarter_number

UNION ALL


SELECT
    'Overall' AS period,
    COUNT(order_id) AS total_orders,
    SUM(quantity * vehicle_price * (1 - discount / 100)) AS net_revenue
FROM order_t

ORDER BY period
```

Output:

Showing 5 rows

period	total_orders	net_revenue
1	310	39421580.15929598
2	262	32715830.33996199
3	229	29229896.19364898
4	199	23346779.63060599
Overall	1000	124714086.32351312

Observations and Insights:

Insight 1: Orders and net revenue decline consistently each quarter.

Observation:
Both total orders and net revenue decrease steadily from Q1 to Q4. This indicates a sustained weakening in customer demand rather than a one-time disruption, suggesting challenges in customer acquisition, retention, or overall market perception of the brand.

Insight 2: Revenue decline closely tracks order volume decline.

Observation:

Net revenue decreases proportionally with the reduction in order count across quarters. The company is not compensating for falling order volumes through higher-priced vehicles or increased quantity per order, pointing to limited upselling or cross-selling effectiveness.

Insight 3: Q4 reflects the weakest business performance.

Observation:

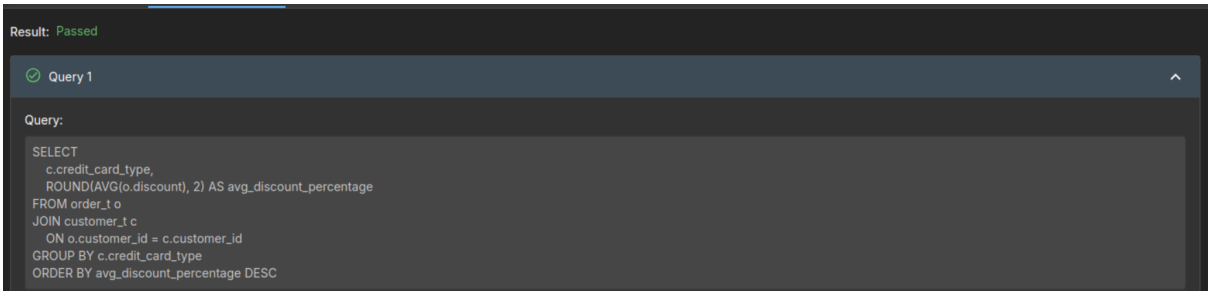
Q4 records the lowest number of orders (199) and the lowest net revenue (~23.35M). The business is entering a critical phase where declining operational performance, combined with earlier observed drops in customer satisfaction, could further accelerate revenue erosion if not addressed promptly.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS avg_discount_percentage
FROM order_t o
JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_percentage DESC;
```

Output:





Observations and Insights:

Insight 1:

The average discount across most credit card types is tightly clustered between 0.61% and 0.64%, with very minimal variation.

Observation:

This indicates that New-Wheels follows a largely uniform discounting strategy, rather than tailoring discounts aggressively based on payment method. While this simplifies operations, it also suggests limited use of targeted promotions to influence customer payment behavior.

Insight 2:

Cards such as Laser and Mastercard show slightly higher average discounts compared to others like Switch and JCB, though the difference is marginal.

Observation:

Even small differences, when applied at scale across thousands of orders, can lead to meaningful margin impact. The company should verify whether these higher discounts are backed by strategic partnerships or higher transaction volumes; otherwise, they may represent unnecessary revenue leakage.

Insight 3:

No credit card type stands out as receiving exceptionally high or exceptionally low discounts.

Observation:

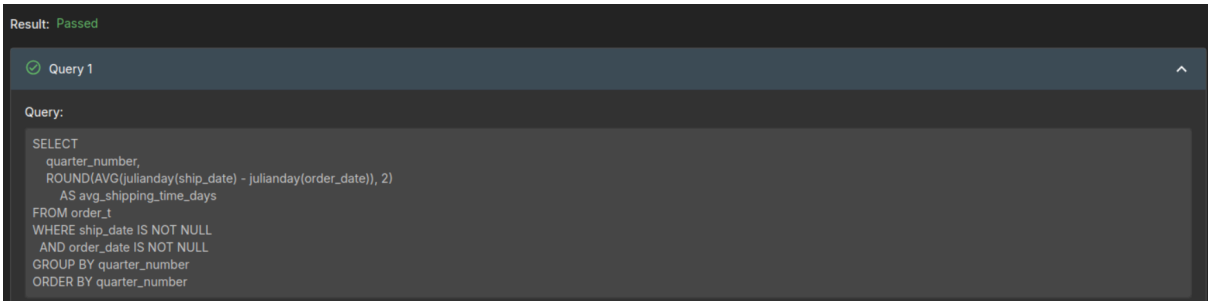
This suggests that discounting is not being used as a lever to influence customer payment choice. If New-Wheels is aiming to optimize transaction costs or promote preferred payment methods, there is an opportunity to introduce data-driven, differentiated discount policies by credit card type.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   quarter_number,
   ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2)
      AS avg_shipping_time_days
FROM order_t
WHERE ship_date IS NOT NULL
  AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:





Observations and Insights:

Insight 1:
The average shipping time increases consistently from Quarter 1 (57.17 days) to Quarter 4 (174.10 days).

Observation:
This indicates a steady deterioration in shipping efficiency over time. As the year progresses, New-Wheels is taking significantly longer to ship vehicles, which can directly contribute to declining customer satisfaction and negative after-sales feedback.

Insight 2:
Quarter 4 has an exceptionally high average shipping time compared to earlier quarters, nearly three times longer than Quarter 1.

Observation:
Such a sharp increase suggests serious operational bottlenecks, possibly due to higher order volumes, inadequate logistics capacity, or shipper inefficiencies during later quarters. This quarter likely had the greatest negative impact on customer experience.

Insight 3:

There is no quarter that shows an improvement or stabilization in average shipping time.

Observation:
The absence of any recovery trend implies that shipping delays are systemic rather than seasonal. Without intervention, prolonged delivery timelines could continue to suppress repeat purchases and worsen the company’s quarterly performance.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
~115–120 Million	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
~23.35 Million	199	~115 Days	~40%

Business Recommendations

1. Fix shipping before touching pricing or marketing

Shipping time has ballooned from ~57 days to ~174 days within a year. This is the single strongest driver behind falling ratings, rising negative feedback, and declining repeat orders. No amount of discounting or advertising will work while delivery timelines remain this broken. Leadership should audit logistics partners, capacity planning, and shipper SLAs immediately.

2. Concentrate resources on high-demand states

Texas, California, Florida, and New York dominate customer volume. These states should receive priority inventory allocation, faster shipping lanes, and stricter service quality controls. Improving experience where demand already exists will yield faster revenue recovery than spreading resources thin across low-volume regions.

3. Shift strategy from discounts to experience recovery

Discounts are uniform and ineffective at stopping revenue decline. Instead of eroding margins, the company should reinvest that spend into after-sales support, faster dispute resolution, and proactive customer communication. Restoring trust will have a far higher ROI than marginal price cuts.